

Konstantinos Christopher Tsiolis

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PROFILE

PhD candidate in Statistics at the University of Toronto and the Vector Institute working on theoretical machine learning, stochastic optimization, high-dimensional learning dynamics, feature learning, and scaling laws.

EDUCATION

Doctor of Philosophy (PhD), Statistics <i>University of Toronto</i> Supervisor: Murat Erdogdu	2023 – 2028 (expected) <i>Toronto, ON</i>
Master of Science (MSc), Mathematics and Statistics <i>McGill University</i> - Supervisor: Adam Oberman - Thesis: Contrastive Learning as Kernel Approximation [Thesis]	2022 – 2023 <i>Montreal, QC</i>
Bachelor of Science (BSc), Mathematics and Computer Science (Honours) <i>McGill University</i> GPA: 3.99/4.0	2018 – 2021 <i>Montreal, QC</i>

PUBLICATIONS AND PREPRINTS

- **Konstantinos Christopher Tsiolis**, Denny Wu, Christos Trampoulidis, and Murat A. Erdogdu. SGD in Multiclass Logistic Regression: Sequential Learning and Scaling Laws. arXiv:2609.07868, 2026. To appear at NeurIPS 2026. [\[Paper\]](#)
- **Konstantinos Christopher Tsiolis**, Alireza Mousavi-Hosseini, and Murat A. Erdogdu. From Information to Generative Exponent: Learning Rate Induces Phase Transitions in SGD. In *Proceedings of the 39th International Conference on Neural Information Processing Systems* (NeurIPS). 2025. [\[Paper\]](#) [\[Code\]](#)
- Jingyi He, **KC Tsiolis**, Kian Kenyon-Dean, and Jackie Chi Kit Cheung. Learning efficient task-specific meta-embeddings with word prisms. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 1229–1241. International Committee on Computational Linguistics, 2020. [\[Paper\]](#) [\[Code\]](#)

ADDITIONAL RESEARCH EXPERIENCE

Research Project on Self-Supervised Learning <i>Mila, McGill University Department of Mathematics and Statistics</i> - Research assistant supervised by Prof. Adam Oberman - Developed and implemented an approach for knowledge distillation of self-supervised models [Report] [Code]	2021 <i>Montreal, QC</i>
Research Projects in Natural Language Processing <i>Mila, McGill Reasoning and Learning Lab</i> - Research assistant supervised by Prof. Jackie Chi Kit Cheung - Implemented and documented the training of word embeddings with various context window sizes [Report] - Developed novel approaches for quantifier scope disambiguation [Report]	2019–2020 <i>Montreal, QC</i>

AWARDS

NSERC Postgraduate Scholarship – Doctoral (PGS D) (120,000 CAD)	2024 – 2027
Faculty of Arts and Science Top (FAST) Doctoral Fellowship (20,000 CAD)	2023 – 2027
Department of Statistical Sciences Student Leadership Award (1,000 CAD)	2025
Queen Elizabeth II Graduate Scholarship in Science and Technology (QEII-GSST) (15,000 CAD)	2023
FRQNT Master’s (B1X) Research Scholarship (17,500 CAD) Application ranked 1st out of 17 in Mathematics category	2022
NSERC Canada Graduate Scholarships - Master’s Program (CGS M) (17,500 CAD)	2022 – 2023
IVADO Undergraduate Introduction to Research Scholarship (6,500 CAD)	2021
NSERC Undergraduate Student Research Award (4,500 CAD)	2020

LEADERSHIP AND SERVICE

Conference Organizer and Reviewer

- Reviewer for *NeurIPS* (2026), *COLT* (2026), *Workshop on High Dimensional Learning Dynamics* (2023 – 2026)
- Member of the Local Organizing Committee for *The Fast and Curious 2: MCMC in Action* (2025)

Mentorship

- **STA492, University of Toronto (Fall 2026)**: Mentoring undergraduate students in AI-assisted research development. Funded by a Pedagogical Innovation and Experimentation grant, Department of Statistical Sciences.
- **University of Toronto Statistical Sciences Directed Reading Program (Winter 2026)**: Weekly meetings with 4 undergraduate students covering graduate-level readings on optimal transport, mean-field neural networks, and lazy vs. rich learning regimes
- **Academic Lecturer at the McGill Artificial Intelligence Society (2020 – 2021)**: Giving 4 two-hour lectures for a student-run machine learning course

President of the University of Toronto Statistics Graduate Student Union (SGSU)

- Elected representative of the PhD and MSc students in the department for the 2024 – 2025 academic year
- Lead organizer of Graduate Student Research Day

TEACHING EXPERIENCE

Teaching Assistant

- **At the University of Toronto (2023 – Present)**: STA2311 and STA2312 (Advanced Computational Methods for Statistics I and II, *PhD Course*), STA347 (Probability), STA314 (Statistical Methods for Machine Learning I), STA261 (Probability and Statistics II), STA221 (The Practice of Statistics II), STA130 (An Introduction to Statistical Reasoning and Data Science)
- **At McGill University (2021 – 2022)**: MATH 222 (Calculus 3), COMP 251 (Algorithms and Data Structures)

SKILLS

Programming Languages: Python, R, Java, Bash, C, MATLAB, JavaScript, OCaml

Libraries: PyTorch, JAX, Hugging Face, NumPy, scikit-learn, pandas, Matplotlib, tidyverse

Tools: Git, LaTeX, Jupyter, VS Code